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Link do repositório:
https://github.com/felipe-uehara/psi3422-lab-sis-elet-2026/tree/main/Aula2_nRF24L01

Vídeo do Funcionamento: https://youtu.be/8f_IDb8M1uA

Código:

```
#include "mbed.h"
#include "nRF24L01P.h"

// The nRF24L01+ supports transfers from 1 to 32 bytes (some boards may
not handle many bytes)
#define TRANSFER_SIZE 1

#define WAIT_TIME_MS 500 // macro for generic delay time, in
milliseconds

/**/ Pins definitions/**/
#define MOSI PTD2 //D11
#define MISO PTD3 //D12
#define SCK PTC5
#define CS PTD0 //D10
#define CE PTD5 //D9
#define IRQ PTA13 //D8

DigitalOut red(LED_RED);
DigitalOut blue(LED_BLUE);

nRF24L01P my_nrf24l01p(MOSI, MISO, SCK, CS, CE, IRQ); // module
instance - FRDM KL25Z
// nRF24L01P my_nrf24l01p(D11, D12, D4, D10, D9, D8); // module
instance (Arduino pinout - compatible to other boards)

/* Objeto Serial */
static BufferedSerial pc(USBTX, USBRX, 9600);

int main() {
```

```

char txData[TRANSFER_SIZE], rxData[TRANSFER_SIZE];
int txDataCnt = 0;
int rxDataCnt = 0;

my_nrf24l01p.powerUp();

// Display the (default) setup of the nRF24L01+ chip
printf( "nRF24L01+ Frequency      : %d MHz\r\n",
my_nrf24l01p.getRfFrequency() );
printf( "nRF24L01+ Output power   : %d dBm\r\n",
my_nrf24l01p.getRfOutputPower() );
printf( "nRF24L01+ Data Rate       : %d kbps\r\n",
my_nrf24l01p.getAirDataRate() );
printf( "nRF24L01+ TX Address        : 0x%01011X\r\n",
my_nrf24l01p.getTxAddress() );
printf( "nRF24L01+ RX Address        : 0x%01011X\r\n",
my_nrf24l01p.getRxAddress() );

printf( "Type keys to test transfers:\r\n (transfers are grouped
into %d characters)\r\n", TRANSFER_SIZE );

my_nrf24l01p.setTransferSize( TRANSFER_SIZE );

my_nrf24l01p.setReceiveMode();
my_nrf24l01p.enable();

while (true)
{

    // If we've received anything over the host serial link...
    if ( pc.readable() ) {

        // ...add it to the transmit buffer
        // txData[txDataCnt++] = getc();
        pc.read(txData,TRANSFER_SIZE);
        txDataCnt++;

        // If the transmit buffer is full

```

```

        if ( txDataCnt >= sizeof( txData ) ) {

            // Send the transmitbuffer via the nRF24L01+
            my_nrf24l01p.write( NRF24L01P_PIPE_P0, txData,
txDataCnt );

            txDataCnt = 0;
        }

        // Toggle LED1 (to help debug Host -> nRF24L01+
communication)
        red = !red;
    }

    // If we've received anything in the nRF24L01+...
    if ( my_nrf24l01p.readable() ) {

        // ...read the data into the receive buffer
        rxDataCnt = my_nrf24l01p.read( NRF24L01P_PIPE_P0, rxData,
sizeof( rxData ) );

        // Display the receive buffer contents via the host serial
link
        for ( int i = 0; rxDataCnt > 0; rxDataCnt--, i++ ) {
            printf("%c",rxData[i]); // putc( rxData[i] );
        }

        // Toggle LED2 (to help debug nRF24L01+ -> Host
communication)
        blue = !blue;
    }
}

```